

Let $\delta_i = \nabla_{Wv_i} \mathcal{L} \in \mathbb{R}^d$ and suppose that W is a simple linear layer. Then, the backpropagation gives

$$G_i = \delta_i v_i^\top$$

We first formalize the assumption that alignment has a sensitive subspace in the representation space where only a subspace of the representation critically impacts alignment. This assumption builds upon most of the existing works claiming the low-rankness of alignment [3, 34, 51, 60, 65].

Assumption 2 (Alignment has a sensitive subspace). *The forward activations $\{Wv_i\}_{i=1}^N$ span a low-rank subspace $\mathcal{U} \subset \mathbb{R}^d$ of dimension $r \ll d$. Furthermore, we assume the error signals δ_i lie within this same subspace $\mathcal{U}$.*

From this assumption, we proceed to show that the average gradient, $\bar{G}$, is therefore low-rank.

Lemma A.7 (Average gradient is low rank). *Let $\bar{G} = \frac{1}{N} \sum_{i=1}^N G_i$ be the average gradient matrix. Under Assumption 1, $\text{Rank}(\bar{G}) \leq r$.*

Proof. By definition, $\bar{G} = \frac{1}{N} \sum_{i=1}^N \delta_i v_i^\top$. Since $\delta_i \in \mathcal{U}$ for all i , we can express each δ_i as a linear combination of the basis vectors of $\mathcal{U}$, denoted $\{u_1, \dots, u_r\}$.

$$\delta_i = \sum_{k=1}^r c_{i,k} u_k$$

Substituting this into the sum:

$$\bar{G} = \frac{1}{N} \sum_{i=1}^N \left(\sum_{k=1}^r c_{i,k} u_k \right) v_i^\top = \sum_{k=1}^r u_k \left(\frac{1}{N} \sum_{i=1}^N c_{i,k} v_i^\top \right)$$

Let $z_k^\top = \frac{1}{N} \sum_{i=1}^N c_{i,k} v_i^\top$. Then $\bar{G} = \sum_{k=1}^r u_k z_k^\top$. This is a sum of r rank-1 matrices. By the subadditivity of rank, $\text{Rank}(\bar{G}) \leq r$. $\square$

We now define the *empirical Fisher Information Matrix*, $\mathbf{F}(S) \in \mathbb{R}^{d^2 \times d^2}$, as the second moment of the flattened gradients. Let $g_i = \text{flatten}(G_i) \in \mathbb{R}^{d^2}$, which gives

$$\mathbf{F}(S) := \frac{1}{N} \sum_{i=1}^N g_i g_i^\top$$

Theorem A.8 (Low Rank Structure of FIM). *Under Assumption 2, the rank of the empirical Fisher Information Matrix $\mathbf{F}(S)$ is bounded by $r \cdot d$. Given that $r \ll d$, $\mathbf{F}(S)$ is low-rank relative to the parameter space dimension d^2 .*

Proof. Let $\text{vec}(\cdot)$ denote the column-wise vectorization of a matrix. The flattened per-sample gradient is obtained by flattening the outer product of δ_i and v_i . This is equivalent to take the Kronecker product of δ_i and v_i .

$$g_i = \text{vec}(\delta_i v_i^\top) = \delta_i \otimes v_i$$

The FIM can be rewritten as:

$$\mathbf{F}(S) = \frac{1}{N} \sum_{i=1}^N (\delta_i \otimes v_i)(\delta_i \otimes v_i)^\top$$

Recall from Assumption 2 that $\delta_i \in \mathcal{U}$, where $\dim(\mathcal{U}) = r$. Let $U \in \mathbb{R}^{d \times r}$ be the basis matrix for $\mathcal{U}$. Then for every i , there exists a coefficient vector $\alpha_i \in \mathbb{R}^r$ such that $\delta_i = U\alpha_i$. We substitute this structure into the vector g_i to obtain

$$g_i = (U\alpha_i) \otimes v_i = (U\alpha_i) \otimes (I_d v_i)$$

Using the mixed-product property of Kronecker products $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, we can factor out the basis matrix:

$$g_i = (U \otimes I_d)(\alpha_i \otimes v_i)$$

The matrix $\mathcal{P} = (U \otimes I_d)$ has dimensions $d^2 \times d \cdot r$. Thus, the span of g_i is constrained by a subspace of dimension $d \cdot r$. Lastly, observe that for any $x \in \dots$, we have

$$\mathbf{F}x = \frac{1}{N} \sum_{i=1}^N g_i (g_i^\top x)$$

is a linear combination of the g_i . Therefore, $\mathbf{F}x$ is in the span of the $\{g_i\}_{i=1}^N$. This gives the final result

$$\text{Rank}(\mathbf{F}(S)) \leq d \cdot r$$

and since $r \ll d$, we have $d \cdot r \ll d^2$, proving $\mathbf{F}(S)$ is low-rank. $\square$

B Non-Static Fisher Information

In the main text, our simplified analysis is carried out with respect to $\mathbf{F}_i(\theta^*)$ and the associated sensitive subspace $M_i(\theta^*)$. Here, we show that the results extend to the setting where $\mathbf{F}_i(\theta)$ varies along the fine-tuning trajectory and the sensitive subspace rotates accordingly. This extension introduces a mild constant-factor and lower-order corrections, and further does not change the qualitative scaling laws established in Section 6.

We impose a mild regularity condition on the Fisher information along $\theta(t)$.

Assumption 3. Fix a skill S_i . Assume there exists constants $r > 0$ and $L_F \geq 0$ such that for all θ which satisfy $\|\theta - \theta^*\| \leq r$, we have that $\mathbf{F}_i(\theta)$ is locally Lipschitz:

$$\|\mathbf{F}_i(\theta) - \mathbf{F}_i(\theta^*)\|_{op} \leq L_F \|\theta - \theta^*\|.$$

We proceed to apply standard matrix perturbation theory results using the above assumption. We define $\Lambda := \lambda_d(\mathbf{F}_i(\theta^*)) - \lambda_{d+1}(\mathbf{F}_i(\theta^*))$, the gap between the d and $d + 1$ largest eigenvalues of $\mathbf{F}_i(\theta^*)$.

The following proposition bounds the rotation of the sensitive subspace.

Proposition B.1 (Davis-Kahan Bound). Under Assumption 3, for all θ with $\|\theta - \theta^*\| \leq r$,

$$\|P_i(\theta - \theta^*)\| \leq \frac{2L_F}{\Lambda} \|\theta - \theta^*\|$$

Proof. This is a direct application of the [17] Theorem for symmetric matrices, reprinted below.

Lemma B.2 (Davis-Kahan sin Θ Theorem). Let $A, B \in \mathbb{R}^{p \times p}$ be symmetric matrices. Let P_A and P_B denote the orthogonal projectors onto the top- k eigenspaces of A and B , respectively. Assume that A has an eigengap at rank k , i.e.,

$$\Delta := \lambda_k(A) - \lambda_{k+1}(A) > 0,$$

where $\lambda_1 \geq \dots \geq \lambda_p$ are the eigenvalues of A . Then

$$\|P_B - P_A\|_{op} \leq \frac{2\|B - A\|_{op}}{\Delta}.$$

Applying Lemma B.2 with $A = \mathbf{F}_i(\theta^*)$ and $B = \mathbf{F}_i(\theta)$ yields

$$\|P_i(\theta) - P_i(\theta^*)\|_{op} \leq \frac{2\|\mathbf{F}_i(\theta) - \mathbf{F}_i(\theta^*)\|_{op}}{\Lambda}$$

which, in combination with the locally Lipschitz assumption, completes the proof. $\square$

Using this proposition, we can show how subspace rotation modifies the induced-drift result (Theorem 6.2).

Theorem B.3. Assume the conditions of Theorem 6.2 and Assumption 3. Let $\theta(t)$ follow gradient flow with $\theta(0) = \theta^*$, and assume $\|\theta(t) - \theta^*\| \leq r$ for $t \in [0, t_0]$. Then, there exists a constant $\rho \geq 0$ such that for all sufficiently small t ,

$$\|\mathbf{F}_i(\theta^*)^{1/2} P_i(\theta(t))(\theta(t) - \theta^*)\| \geq \frac{\gamma}{2} t^2 - \sqrt{\lambda_{\max}} \epsilon t - \rho t^2 - O(t^3) \quad (13)$$

Proof. Denote $P(t) := P_i(\theta(t))$, $\mathbf{F} := \mathbf{F}_i(\theta^*)$, and $\Delta\theta(t) := \theta(t) - \theta^*$ for readability. By the triangle inequality,

$$\|\mathbf{F}^{1/2} P(t) \Delta\theta(t)\| \geq \|\mathbf{F}^{1/2} P(0) \Delta\theta(t)\| - \|\mathbf{F}^{1/2} (P(t) - P(0)) \Delta\theta(t)\|,$$

By Theorem 6.2, we can directly bound the first term from below. For the second term, we invoke Proposition B.1

$$\|\mathbf{F}^{1/2} (P(t) - P(0)) \Delta\theta(t)\| \leq \|\mathbf{F}^{1/2}\|_{\text{op}} \|P(t) - P(0)\|_{\text{op}} \|\Delta\theta(t)\|.$$

Collecting terms yields the inequality. $\square$

Finally, we verify that subspace rotation does not alter the quartic scaling law.

Corollary B.4. Under the conditions of the above theorem and Theorem 6.1,

$$\Delta u_i(\theta(t)) = \Omega(t^4)$$

for sufficiently small t , provided $\gamma > \rho$.

Proof. Applying Theorem 6.1 with $P_i(\theta(t))$ and substituting the lower bound from the above theorem yields

$$\Delta u_i(\theta(t)) \geq \frac{1}{2} \|\mathbf{F}^{1/2} P_i(\theta(t)) \Delta\theta(t)\|^2 - O(t^3)$$

Since the leading term grows quadratically in t , we obtain the desired quartic growth. $\square$

C Relaxation of Assumption 1

We begin by isolating the instances where Assumption 1 is used in the main results. Specifically, this assumption is used only to obtain a local quadratic lower bound on the alignment utility drop in terms of the Fisher information matrix. This motivates the following weaker (yet more complicated) condition.

Assumption 4 (Local Fisher lower bound for skill utility). For each alignment skill S_i , there exist constants $r_i > 0$ and $C_i \geq 0$ such that for all $\Delta\theta$ with $\|\Delta\theta\| \leq r_i$,

$$\begin{aligned} \Delta u_i(\theta^* + \Delta\theta) &:= u_i(\theta^*) - u_i(\theta^* + \Delta\theta) \\ &\geq \frac{1}{2} \Delta\theta^\top F_i(\theta^*) \Delta\theta - C_i \|\Delta\theta\|^3. \end{aligned}$$

The original assumption directly implies these conditions.

Lemma C.1 (Realizability implies local Fisher lower bound). Suppose Assumption 1 holds and that $\pi_\theta(y|x)$ is three-times continuously differentiable in θ in a neighborhood of θ^* , with bounded third derivatives. Then Assumption 4 holds with

$$F_i(\theta^*) = \mathbb{E}_{x \sim \mathcal{D}_i} \mathbb{E}_{y \sim \pi_{\theta^*}(\cdot|x)} \left[\nabla_\theta \log \pi_\theta(y|x) \nabla_\theta \log \pi_\theta(y|x)^\top \right]_{\theta=\theta^*}.$$

Lastly, this condition is sufficient to obtain our main theorems.

Lemma C.2 (Sufficiency of Assumption 4). Assume Assumption 4 holds for skill S_i . Then Theorem 6.1, Theorem 6.2, and Corollary 6.3 all remain valid with unchanged statements, up to constant factors.

Proof. Theorem 6.1 is precisely our Assumption 4 rewritten in Fisher norm form. Moreover, the results of Theorem 5.2 rely only on the AIC, thus we have the result. $\square$